Mehrad **Haghshenas**

PhD, University of Waterloo

Canada

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Research Interests

Formal Verification | **Programming Languages** | **Compilers**
Automated Theorem Proving | **Static Analysis** | **Functional Programming**

Education

University of Waterloo *Canada*

Ph.D. in Computer Science (member of the CrySP lab) *Expected: Jan. 2029*

- Cumulative GPA: **94.00 / 100.00**.
- Researching semantics-directed conformance testing of language interpreters through SMT-based program synthesis. A language specification is encoded as an executable specification in a Rust-embedded DSL, whose semantics are automatically lifted into SMT-LIB2 constraints. By targeting individual error paths, Z3 synthesizes inputs guaranteed to exercise those paths, where each input encodes a semantic obligation that any conforming implementation must satisfy. This enables systematic detection of specification violations in commercial interpreters. Supervised by Dr. Meng Xu.

Utrecht University *Netherlands*

Computing Science *Graduation: Jan. 2024*

- Cumulative GPA: **3.94 / 4.00** *(Summa cum laude)*.
- Thesis: *Automated Verification of Hoare Triples: A Comparative Study on Static Inference of Loop Invariants* – Grade: **4.00 / 4.00**, supervised by Dr. S.W.B. Wishnu Prasetya and Prof. dr. ir. Frank van der Stappen.

Awards

2025 **International Doctoral Student Award (IDSA)**, Awarded by the *University of Waterloo*. *Canada*

2023 **Scholarship**, Awarded by *Utrecht University*. *Netherlands*

Work Experience

Shopify *Canada*

Software Developer *Jan. 2026 - May 2026*

- *Worked on the agentic e-commerce project using Ruby on Rails in the CT&I team.*

ABN AMRO *Netherlands*

Full-time Mainframe Developer *Mar. 2024 - Dec. 2024*

- *Reviewed and refactored legacy COBOL code. Investigated automated testing frameworks (Topaz for Total Test). Updated technical documentation. Collaborated in Agile teams following Scrum methodologies.*

Hudson Dynamics B.V. *Netherlands*

Embedded Software Engineer Intern *Jun. 2023 - Dec. 2023*

- *Worked on the development of automated vehicles. Programmed the robotic arm of an electric loader to ensure precise vertical lift path control. Designed its control system using Parker IQAN software. Modeled the arm's search space with Denavit–Hartenberg notation and applied forward and inverse kinematic analysis.*

Volunteering Activities

Women in Mathematics Program *University of Waterloo*

Mentor *Winter 2026*

- Defined and supervised a research project on formally verifying SAT encodings in Coq, bridging proof assistants with SAT/SMT-based reasoning.
- Recipient of the *Women in Mathematics Mentorship Award* in recognition of mentorship contributions.

Artifact Evaluation Committee *PoPETs*

Program Committee Member *Oct. 2025*

- Served on the committee for the Proceedings on Privacy Enhancing Technologies symposium, reviewing the *ReporTor: Facilitating User Reporting of Issues Encountered in Naturalistic Web Browsing via the Tor Browser* artifact to ensure reproducibility.
- Contributed to the open science mission by providing feedback to authors on artifact quality, completeness, and reusability.

Hack the North *University of Waterloo*

Participant *12 – 14 Sep. 2025*

- We developed *CodeMind*, a local-first, searchable timeline tool that documents AI coding sessions. The tool uses a custom Model Context Protocol (MCP) server to ingest sessions from Cursor/Claude, and an OpenAI-based LLM to extract and index key information.

Rust Z3 bindings
GitHub

OPEN-SOURCE CONTRIBUTOR TO THE Z3 RUST API. *Sep. 2025*

- Contributed feature enhancements including floating-point NaN constructors, Int–Real mixed arithmetic, 128-bit integer support, string comparison, sequence utilities, and rounding-mode APIs. Implemented correctness fixes such as improved handling of *Probe::It* arguments and model retrieval documentation. Authored the Rust bindings guide, published on the official Z3 project website.

Stanford University Code in Place
Online

VOLUNTEER TUTOR *Apr.–May 2023, 2024, & 2025*

- Volunteer tutor for Stanford's Code in Place (2023, 2024, 2025), mentoring 15 students over 6 weeks each year.

Formal Methods Europe
Europe

MEMBER *Oct. 2023 – Present*

- An organization that promotes research in formal methods to improve software and hardware systems.

Utrecht University Hackathon
Netherlands

PARTICIPANT *24 – 25 May 2023*

- Co-hosted by SUE Co., the event focused on the applications of AI in Software Development, GitHub Copilot, and AWS.

TEDx Organizing Committee
Netherlands

ORGANIZING COMMITTEE VOLUNTEER, TEDX EVENT. *15 Apr. 2023*

- Served on the organizing committee for a TEDx event focused on the themes Fight or Flight and Sustainability.

Summer School

Queen Mary University of London
England

THREE-WEEK COURSE: *Business and Society: The Changing World of Work* *Jul. 2022–Aug. 2022*

- 15 UK credits (7.5 ECTS); grade: **A** (75.6/100).

Utrecht University
Netherlands

ONE-WEEK COURSE ON *Advanced Functional Programming in Haskell* (DR. WOUTER SWIERSTRA) *Jul. 2022*

- No exam was administered.

Miscellaneous

Languages English (**IELTS 8.5**, taken on Nov. 3, 2023), Persian (native), French (A1).

Programming Rust, Coq, C, Python, Haskell, Java, COBOL, MATLAB, JavaScript, R, SQL, C#.

Relevant Courses Software & Systems Security, Languages & Compilers, Functional Programming, Software Testing & Verification, Logic for Computer Science, Operating Systems, Software Verification Using Proof Assistants.